



Google Pay to add gold gifting

Google is planning to add gold gifting options to its India-only payment app in addition to a theme redesign. <https://digit.in/dec19-41>



Malicious mobile SDKs

Facebook and Twitter are warning of SDKs that can scrape and share personal information. <https://digit.in/dec19-42>

SOFTWARE MACROS

Life's been made a lot simpler with the help of macros, here's how it came to be

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The concept of software macros dates back to the 1950s when assembly language programming was used to create software programs. The purpose of software macros is to reduce lengthy coding by creating predefined building blocks with a specific set of instructions as a single chunk of code or program. You can plugin the same macro in several software applications without having to repeat lengthy code. Let's delve into the origins of software macros.

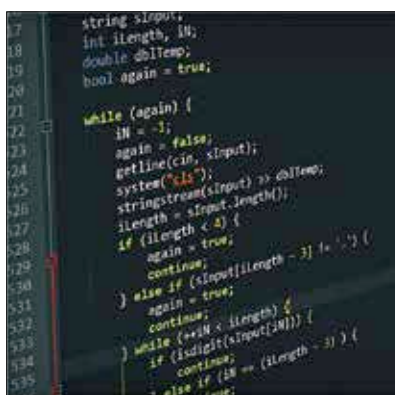
ORIGIN

The term 'macro' has existed since the conception of programming languages and it is used to automate common tasks. Macros were initially developed using the IBM 705 computer for the Dow Chemical Corporation and Air Material Command. In 1963, Timothy Hart proposed the addition of macros to LISP 1.5 in AI Memo 57. A macro instruction coded as per the framework of the target assembly language will be processed by a macro compiler. The macro language was followed by Macro Assemblers in the late 1950s. Programmers wrote a macro library for every target machine but not the complete assembly language program. The arrival of VBA and the

ability to create macros from within Microsoft Office set new benchmarks in the software development arena starting from the late 1990's.

Keyboard and mouse macros

A keyboard macro combines a sequence of tedious keystrokes into one simple program and it dates back to early 80s. You can club commonly used key strokes into one single macro unit. You can also automate frequently used mouse commands into single macro. It is possible to build macros using various third-party



Macros enhances the productivity of programmers by avoiding repetitive tasks

tools. Some of the recorders available during the early 1980s are SmartKey, SuperKey, KeyWorks and Prokey. In the last few years, keyboard macros have been the primary source of exploitation when it comes to the Massively Multiplayer Online Role Playing Games (MMORPGs). The usage of macros in online gaming enabled users to earn plenty of points illegally.

Nowadays, keyboard and mouse macros are created using the integrated macro module included with the relevant Integrated Development Environment aka IDE. We still remember how

we worked with macros from within Visual Basic 6 in the early 2000s. Text editors such as Emacs and Vim were developed using macros. Emacs was used along with TECO followed by LISP. The Vim editor provided the capability to record keyboard actions. You could also create macros using Vimscript and Visual Basic for Applications (VBA) included with Office 97 through Office 2019.

Visual Basic for Applications

The purpose of VBA was to enable developers to build user-defined functions including automation of processes and was primarily used with Microsoft Excel. It could also access the Windows API and other functionalities via Dynamic Link Library (DLL). VBA basically expanded the capabilities of WordBasic, which was application-specific macro programming languages. You can manage various aspects of the host application in addition to manipulation of user interface features like Menus and Toolbars.

Microsoft has included VBA with Microsoft Office, Office for Mac OS X, Microsoft MapPoint and Microsoft Visio. Some other notable applications with VBA that are not developed by Microsoft are CorelDraw, LibreOffice, AutoCAD, and WordPerfect among others. The problem with VBA is that hackers made use of the system to create computer viruses and malwares. However, the macros virus issue has been slowly addressed by the manufacturers during the development phase of the anti-virus programs.

Emacs

Emacs, also called Editor Macros, is a text editor with over 10000 built-in commands and was developed using macros. The first edition of the Emacs was released in 1976 by Carl Mikkelsen, David A. Moon and Guy L. Steele Jr for the TECO Editor drawing inspiration from TECMAC and TMACS. The GNU Emacs, which is the ported version of Emacs was developed by Richard Stallman. In 1991, XEmacs was carved out from GNU Emacs. As a developer, you